Junehyoung Jeon

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Research Interests

I am interested in how the early Universe and its components such as galaxies, first stars, and black holes evolved. I approach these problems creating/using theoretical models and hydrodynamical simulations.

Education

Doctor of Philosophy in Astronomy — Expected completion May 2027
Department of Astronomy, University of Texas at Austin, Austin, Texas

Master of Arts in Astronomy — August 2023
Department of Astronomy, University of Texas at Austin, Austin, Texas

Bachelor of Science in Astrophysics, Minor in Physics — May 2021
Barrett Honors College, Arizona State University, Tempe, Arizona
Average unweighted GPA: 4.0/4.0

Publications

First-Author Publications

Jeon, J. et al. (2026). *Early Supermassive Black Holes and Little Red Dots Require Free-Fall Growth.* arXiv:2609.00115, Submitted to the Open Journal of Astrophysics.

Jeon, J. et al. (2026). *What is Powering the Enigmatic He* II *Emitter Hebe: The First Stars or Black Holes?.* The Astrophysical Journal, Volume **1006**, Issue 1, 27.

Jeon, J. et al. (2026). *Hunting for the First Explosions at the High-Redshift Frontier.* The Astrophysical Journal, Volume **1001**, Issue 1, 3.

Jeon, J. et al. (2026). *Little Red Dots and their Progenitors from Direct Collapse Black Holes.* The Astrophysical Journal, Volume **998**, Issue 1, 148.

Jeon, J. et al. (2025). *The Emerging Black Hole Mass Function in the High-Redshift Universe.* The Astrophysical Journal, Volume **988**, Issue 1, 110.

Jeon, J. et al. (2025). *Physical Pathways for JWST-Observed Supermassive Black Holes in the Early Universe.* The Astrophysical Journal, Volume **979**, Issue 2, 127.

Jeon, J. et al. (2023). *Observability of Low-Luminosity AGN in the Early Universe with JWST.* Monthly Notices of the Royal Astronomical Society, **524**, 176-187.

Jeon, J. et al. (2022). *Maximal X-ray feedback in the pre-reionization Universe.* Monthly Notices of the Royal Astronomical Society, **515**, 5568-5575.

Co-Author Publications

Zhang, S., **Jeon, J.**, et al. (2026). *The Road to Normalcy: Environment-Driven Evolutionary Pathways for Primordial Black Holes.* arxiv:2608.23498, Submitted to ApJ.

Gupta, A. et al. (**Incl. Jeon**) (2026). *A Rapid Evolution in the Observed $M_{\rm BH}/M_*$ Relation at $z > 3$ Revealed via Spectro-photometric SED-Modeling.* arxiv:2605.30414, Submitted to ApJ.

Fei, Q. et al. (**Incl. Jeon**) (2026). *A GLIMPSE of Intermediate Mass Black holes in the epoch of reionization: Witnessing the Descendants of Direct Collapse?.* The Astrophysical Journal, Volume **1003**, Issue 2, 244.

Zhang, S., Liu, B., Bromm, V., **Jeon, J.**, et al. (2025). *How do Massive Primordial Black Holes Impact the Formation of the First Stars and Galaxies?*. The Astrophysical Journal, Volume **987**, Issue 2, 185.

Taylor, A., Finkelstein, S., Kocevski, D., **Jeon, J.**, et al. (2025). *Broad-Line AGN at* $3.5 < z < 6$*: The Black Hole Mass Function and a Connection with Little Red Dots*. The Astrophysical Journal, Volume **986**, Issue 2, 165.

Invited Talks

Heavy Seeds and the Multi-scale Fueling of Early Supermassive Black Holes and Little Red Dots
Theoretical Astroparticle and Cosmology Symposium, Texas A&M University — October 2026

Conference and Seminar Talks

The Emerging Black Hole Mass Function in the High-Redshift Universe
Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman, Pasadena, United States — July 2026
Cosmic Frontier Center Inaugural Conference, Austin, United States — May 2025

Little Red Dots and their Progenitors from Direct Collapse Black Holes
Little Red Dots 2026, Online Workshop — June 2026
ITC Lunch, Harvard Center for Astrophysics — September 2025

Hunting for the First Explosions at the High-Redshift Frontier
Spring Symposium, Space Telescope Science Institute — May 2026

Formation and Evolution of Supermassive Black Holes in the Early Universe
Galaxy Formation Seminar, Center for Computational Astrophysics — October 2025
Seminar, Columbia University Astronomy & Astrophysics — October 2025
MAT talks, MIT Kavli Institute — September 2025
HEAD Seminar, Harvard Center for Astrophysics — September 2025
Galaxy and AGN Journal Club, Space Telescope Science Institute — September 2025

Physical Pathways for JWST-Observed Supermassive Black Holes in the Early Universe
CRISOL25, Toledo, Spain (remote) — May 2025
Probing the Genesis of SMBH, Kashiwa, Japan (remote) — November 2024
CCA-CFC Workshop, Austin, United States — November 2024
ITC Lunch, Harvard Center for Astrophysics — November 2024

Conditions for Efficient Growth of Supermassive Black Holes in the Early Universe
COSPAR 2024, Busan, South Korea — July 2024
Massive Black Holes in the First Billion Years, Kinsale, Ireland — April 2024

Observability of Low-Luminosity AGN in the Early Universe with JWST
Black Holes on Broadway, New York City, United States — December 2023
Young Astronomers on Galactic Nuclei, Palermo, Italy — October 2023

Awards and Honors

Research covered by AAS Nova and Astrobites
College Continuing Graduate Fellowship, UT Austin ($10,000) — 2026
David Alan Benfield Memorial Fellowship in Astronomy, UT Austin ($1,500) — 2025
Graduate School Dissertation Writing Fellowship, UT Austin ($24,000) — 2025
Graduate Summer Excellence Award, UT Austin ($10,000) — 2025
Board of Visitors Graduate Student Endowment Fund ($10,000), UT Austin — 2023

Professional Development Award, UT Austin ($500) 2023
The College of Liberal Arts and Sciences Dean's Medal, ASU 2021
Moeur Award, ASU Alumni Association 2021
New American University Award ($5,000) 2018–2021
The College of Liberal Arts and Sciences Dean's List, ASU 2018–2021

Professional Services

Referee for Physical Review Letters, Physical Review D, and the Astrophysical Journal

Teaching and Mentoring

Teaching Assistant, University of Texas at Austin

- *Astrophysics* Fall 2026, Spring 2025
  - Assisted astronomy and physics major upper class students with assignments and projects, helping their understanding of the lecture concepts and applications.
  - Prepared and taught introductory python modules designed for students who have no prior experience, introducing them to the basics of coding in astrophysics.
- *Practical Introduction to Research* Spring 2024, Fall 2021
  - Guided astronomy and physics major students through in-class activities such as coding, writing resume, and making posters, providing them with introductory skills in academia.
  - Prepared coding and lecture modules for the students, leading the students to understand the concepts.
- *Cosmology* Fall 2022
  - Explained course topics to students after class, helping them individually to grasp lecture concepts that they might not have fully understood.
  - Guided students through their homeworks, assisting them in understanding questions and steps to solve various cosmology problems.

Undergraduate Research Projects, University of Texas at Austin Fall 2024 – Current

- Mentored undergraduate students on research projects on the first supermassive black holes, enabling the students to perform theoretical modeling and analysis.
- Guided the students in writing research publications and conference presentations, producing concrete results from the students' work.

Mentor, Undergraduate Mentorship in Astronomy Program,
University of Texas at Austin Fall 2023 – Current

- Advised multiple undergraduate mentees on various topics, including research, graduate school preparation, and applications, helping them advance their academic career.
- Held weekly meetings to discuss updates and new questions brought up by the mentees, guiding them in general academics.

Skills

Programming: Python, C, Fortran, High Performance Computing
Operating systems: Windows, Linux
Software: LaTeX, Mathematica
Languages: English, Korean